TEXORA · LECTURE SHEET

Fabric Finishing Machines and Agents

The stenter, padding mangle, curing oven, calender, sanforizer and foam line — what each machine does, how it is controlled, and the catalogue of agents it delivers.

WP-4201 · TEXTILE FINISHING · TOPIC 4 · SPECIALIZATION — WET PROCESSING

1. The finishing line as a system

Topics 2 and 3 described effects and chemistries. This sheet describes the hardware that produces them, in the order fabric usually meets it: **padding mangle** (apply liquor) → **stenter** (dry, width-set, heat-set) → **curing oven** (react the finish, where separate curing is used) → **calender / sanforizer / raising–shearing** (mechanical surface and shrinkage work) → inspection and rolling. Foam application can replace the padding mangle for low-liquor work. One machine often does two jobs — a stenter with a padder at entry and high-temperature chambers is simultaneously applicator, dryer and curing unit — so the finisher thinks in *functions*, then assigns them to machines.

2. Stenter (stenter frame)

The stenter is the most important finishing machine. Fabric enters over feed and straightening rollers, its selvedges are pinned (pin stenter) or gripped (clip stenter) onto two endless chains, and the chains diverge to pull the fabric to the finished width while it travels through a series of hot-air chambers; at exit the chains converge slightly, the fabric is released, cooled and batched. **Overfeed** at entry (feeding fabric 5–20% faster than chain speed, by a pinning wheel/overfeed roller) lets the fabric shrink lengthwise, raising GSM and cutting residual warp shrinkage. Air is blown through nozzles above and below the fabric; each chamber ("zone" or bay) has its own temperature, so a drying profile (gentle first, hotter later) and a final high-temperature zone for curing or heat setting can be set on one pass.

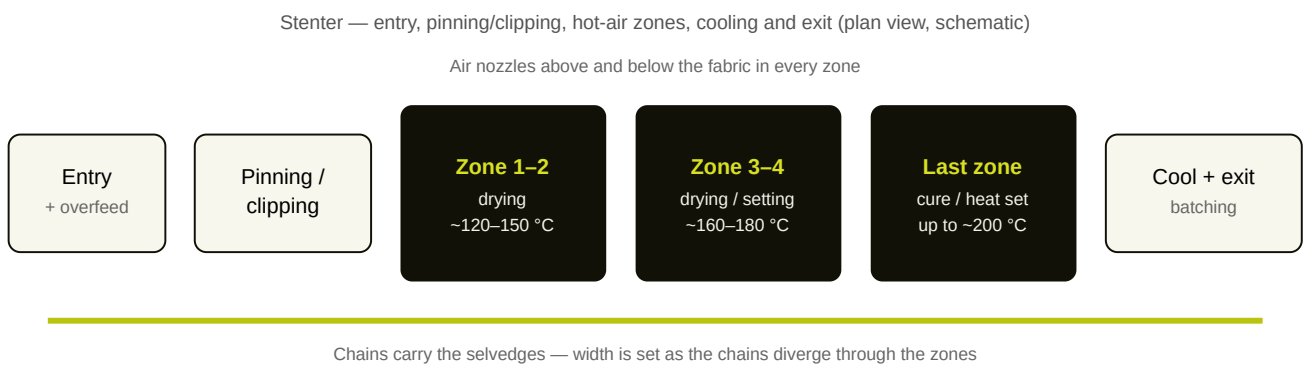


Figure 1 — Stenter schematic: entry with overfeed → pinning/clipping → graduated hot-air zones → cooling and exit.

FEATURE	PIN STENTER	CLIP STENTER
Selvedge held by	Rows of pins impaling the edge	Clips gripping the edge
Best for	Wovens, fabrics needing firm edge control; leaves pin holes at the very edge	Knits and delicate fabrics; no holes, but a small ungripped-to-gripped edge behaviour differs
Common duties	Drying padded finishes, final width setting, heat setting of synthetics, resin curing in high-temperature zones, combined with weft straightener and overfeed control	

Controls and faults: temperature profile per zone, chain speed (dwell time), overfeed %, width sensors, exhaust humidity control. Running faults map directly to quality: bowing/skew from entry straightening, width variation from chain problems, migration and two-sidedness from too-hot early zones, and scorching or disperse-dye sublimation from excessive final-zone temperature.

3. Padding mangle

The padding mangle (padder) applies finishing liquor evenly. Fabric passes through a trough of liquor and then through the nip of two (or three) rollers — commonly a steel roller against rubber-covered rollers, loaded pneumatically or hydraulically — which squeeze out surplus to a controlled **wet pick-up** (Topic 3). Two-dip/two-nip arrangements improve penetration on dense fabrics. Pick-up is set by nip pressure, roller hardness, fabric construction and speed, and is verified gravimetrically: weigh a dry sample, pad it, weigh again. Pick-up % = (padded mass − dry mass)/dry mass × 100. Uneven nip pressure across the width gives side-to-side add-on variation — the first suspect in any side-centre-side finishing complaint.

WORKED EXAMPLE — PICK-UP CHECK

A dry fabric sample weighs 200.0 g. After padding in the resin bath it weighs 340.0 g.
Pick-up = (340.0 − 200.0)/200.0 × 100 = **70%** — exactly the figure used in Topic 3, so the

> DMDHEU 60 g/L bath delivers 42 g/kg (4.2% owf) on this setting. If the padded mass were only 300.0 g (50% pick-up), the same bath would deliver just 30 g/kg, and crease recovery would fall short although "the recipe" never changed.

4. Curing oven

Where curing is not done in the stenter's last zones, a separate **curing oven** (hot-flue / loop or roller curing range) holds the dried fabric at reaction temperature for a timed dwell: resin curing typically 150–170 °C for 1–3 min equivalent. The engineering requirements are uniform temperature across width and length (a cold lane leaves uncured, formaldehyde-releasing, wash-unstable finish), accurate dwell (speed × chamber length), and exhaust of volatiles. Thermosol and heat-setting duties overlap with the stenter; the choice between stenter-cure and separate cure is capacity and fabric-sensitivity driven. Temperature is verified with temperature-indicating strips/papers run through on the fabric, not trusted to the panel alone.

5. Calender machine

Construction was drawn in Topic 2 (Figure 1 there): a stack of 2–7 bowls, alternating heated steel and resilient (cotton, paper or polymer) bowls, loaded by hydraulic pressure, driven so that bowls may run at equal surface speed (swizzing) or with a speed difference (friction). Embossing and Schreiner work substitute an engraved steel bowl. Machine variables: nip pressure/loading, bowl temperature, speed, number of passes and fabric moisture — a slightly damp fabric calenders to a higher lustre than a bone-dry one. Safety and quality points: nip guarding, even loading to avoid one-sided lustre, and bowl condition (a scored resilient bowl prints its defect every revolution).

6. Sanforizing machine — rubber-belt unit

The full sanforizing range (schematic in Topic 2, Figure 2) runs: entry and moistening (steam or spray box brings the fabric to a controlled damp state) → optional cloth/blanket pre-shrink or tensionless drying stage → the **rubber-belt unit** (endless thick rubber belt, feed roller that stretches the belt surface, pressure/heated cylinder against which the belt contracts — the shrinkage zone) → **Palmer drying unit** (fabric held on a felt blanket against a large steam-heated cylinder, dried without tension so the imposed shrinkage is not pulled back out) → cooling, inspection and batching. Belt condition is the machine's heart: a worn, cracked or unevenly ground belt gives uneven shrinkage across the width, checked by wash-testing left, centre and right specimens (target residual ≤1% for a sanforized claim, as calculated in Topic 2: 50.0 cm → 49.6 cm = 0.8%).

7. Raising and shearing machines

- **Raising machine:** a large rotating drum (or a series of rollers) covered with many small wire-toothed rollers (pile and counter-pile rollers rotating in opposite senses), each with card-clothing hooks that tease fibre ends from the yarns as fabric passes over the drum. Passes, roller speeds relative to fabric speed, and hook condition set pile height; the fabric may be raised dry, damp or wet according to fibre (wool is often raised damp).
- **Shearing (cropping) machine:** fabric, held flat and taut over a rest, passes under a rapidly rotating cylinder of spiral blades working against a fixed ledger blade, with a brush and suction to remove the cut fibres. Cutting height is set in fractions of a millimetre above the ground fabric — too low scars the cloth, too high leaves an uneven pile after raising.
- **Brushing machine:** rotating bristle/nylon brushes, gentler than either — used for surface cleaning, pile laying and pre-inspection preparation.

8. Foam finishing — generator and applicator basics

In foam finishing the agent liquor is whipped with air in a **foam generator** (liquor + surfactant foaming aid metered into a mixing head with compressed air) to a stable foam of defined **blow ratio** (foam volume : liquor volume, commonly about 5:1 to 20:1). The foam is spread on the fabric by an applicator — a horizontal screen/roller or knife system — and collapses on and in the fabric under a nip or vacuum, depositing the agent with far less water than padding: pick-ups of 10–30% instead of 60–80%. Benefits: large drying-energy saving, less migration, one-sided application is possible, and wet-on-wet processing. Controls: foam density and stability (a foam that collapses too early floods the fabric; too stable, it will not distribute), application weight, and collapse method. Foam is an *application technology*, not a chemistry — resin, repellent and softener recipes of Topic 3 can all be foamed, with concentrations recalculated for the low pick-up.

> **WORKED EXAMPLE — FOAM CONCENTRATION CORRECTION**
>
> Padding delivers DMDHEU at 42 g/kg using 60 g/L at 70% pick-up (Topic 3). Foam applies only 20% pick-up = 0.20 L/kg.
>
> Required foam-liquor concentration = 42 g/kg ÷ 0.20 L/kg = **210 g/L**.
>
> Same add-on, much stronger bath — the standard recalculation whenever the application method changes the pick-up.

9. Finishing agents catalogue

AGENT	CHEMISTRY	FUNCTION	TYPICAL DOSAGE

DMDHEU resin	N-methylol crosslinker for cellulose	Easy care, crease recovery, wash-and-wear	40–80 g/L (pad)
$MgCl_2$ catalyst	Latent acid catalyst (Lewis acid)	Triggers resin crosslinking during cure	~15–25% on resin mass
Cationic softener	Quaternary ammonium compound	Full soft handle (exhaust or pad)	10–30 g/L pad; 1–3% owf exhaust
Silicone softener	Amino-functional PDMS micro-emulsion	Sleek softness, sewability, tear gain	10–30 g/L (20 g/L ⇒ 14 g/kg at 70% pick-up)
Polyethylene emulsion	Non-ionic PE wax emulsion	Tear/abrasion improvement in resin baths	10–40 g/L
Paraffin repellent	Wax + Al/Zr salt	Water repellency, low cost	20–50 g/L
Silicone repellent	Polysiloxane + catalyst	Water repellency with soft handle	10–30 g/L
Fluorocarbon repellent	Fluoropolymer (C6 type)	Water + oil repellency, durable	20–60 g/L
THPC FR system	Phosphonium salt + urea, ammonia cure	Durable flame retardancy, cotton	Per supplier system, ~15–25% owf range
Pyrovatex-type FR	N-methylol phosphonopropionamide	Durable FR by pad-dry-cure with resin	Per supplier system, with resin co-bath
Soil-release polymer	Hydrophilic PEG/acrylic copolymer	Oily-soil release from polyester/blends	20–40 g/L
Antistatic agent	Quaternary / ethoxylated hygroscopic type	Static dissipation on synthetics	5–20 g/L
Antimicrobial	Silver-ion or bonded Si-Quat type	Bacterial/fungal inhibition, odour control	Per supplier; dosed to test standard
Cellulase enzyme	Acid cellulase (biological catalyst)	Bio-polishing, depilling of cellulosics	~0.5–2% owf, exhaust, pH 4.5–5.5
Stiffener (temporary)	Starch / CMC / PVAc emulsion	Body and stiffness, low cost	20–100 g/L by target stiffness

Dosages are starting windows for recipe design, always confirmed by laboratory trials against the buyer's test specification; supplier technical data for the exact commercial product governs in

production.

10. Machine-to-finish cross-reference

FINISH (TOPICS 2–3)	APPLICATION	DRYING / SETTING	FIXING / FINAL MACHINE
Resin easy-care	Padding mangle (or foam)	Stenter	Curing oven or stenter hot zones; then sanforizer
Softening	Padder or exhaust vessel	Stenter	Calender if lustre also specified
Water repellency	Padding mangle (or foam)	Stenter	Cure in stenter/oven for silicone and fluorocarbon
Flame retardant (Pyrovatex type)	Padding mangle	Stenter	Curing oven (with the resin system)
Flame retardant (THPC)	Padding mangle	Dryer	Ammonia curing chamber + oxidation, wash-off
Calender effects	—	Conditioning (moisture)	Calender (swizzing / friction / Schreiner / embossing)
Raised + levelled surface	—	—	Raising machine → shearing machine
Sanforized cotton	Moistening at sanforizer entry	Palmer unit (in range)	Rubber-belt sanforizer itself
Heat-set polyester	—	—	Stenter at 180–200 °C
Bio-polishing	Exhaust in jet/winch	Hydroextract → stenter/compactor	Enzyme deactivation stage in the batch machine

11. Glossary — key terms

- **Stenter:** chain machine holding fabric by the selvedges to dry, set width and heat-set or cure.
- **Wet pick-up:** liquor retained after the padding nip, % of dry mass — the link between g/L and add-on.
- **Dwell time:** time the fabric spends at temperature in stenter or curing oven (chamber length ÷ speed).
- **Palmer unit:** felt-blanket cylinder dryer that dries sanforized fabric without tension.

- **Blow ratio:** volume of foam produced per volume of liquor in foam finishing.
- **Migration:** movement of dissolved/dispersed agent during drying, causing uneven (often two-sided) finish.

12. Key points

- Think functions first: apply → dry → cure/set → mechanical finish; machines share functions (the stenter does three).
- Pick-up is measured by weighing, never assumed — 200.0 g → 340.0 g is 70%, and it fixes the add-on.
- Foam trades water for concentration: 20% pick-up needs 210 g/L to match 42 g/kg add-on.
- Every Topic 2–3 finish maps to a machine train — use the cross-reference table to write a full route.

13. Summary

The finishing department is a small set of versatile machines — padder, stenter, curing oven, calender, sanforizer, raising and shearing machines, and the foam line — combined into a route for each fabric and specification, delivering a catalogue of agents whose dosages are set in g/L, converted by pick-up into add-on, and proved by testing. With Topic 1's classification, Topic 2's mechanical effects and Topic 3's chemistry, this sheet completes WP-4201: given any buyer's finishing specification, the route, recipe, machines and checks can now be written in full.

14. Practice

1. A dry sample of 150.0 g weighs 255.0 g after padding. Calculate the wet pick-up, then the add-on of a softener present at 25 g/L.
2. A stenter has 8 chambers of 3 m each and runs at 30 m/min. Calculate the dwell time in the hot zones, and state whether it suits a 20-second heat set.
3. Write the full machine train (in order) for: (a) resin-finished, sanforized cotton shirting; (b) fluorocarbon-repellent polyester jacket fabric.
4. Foam application at 15% pick-up must deliver a repellent add-on of 12 g/kg. What foam-liquor concentration is required?
5. Left-side specimens of a sanforized fabric shrink 0.8% but right-side specimens shrink 2.3%. Name the likely machine cause and the check you would make.

Answer key

1. Pick-up = (255.0 − 150.0)/150.0 × 100 = 70%. Add-on = 25 × 0.70 = 17.5 g/kg = 1.75% owf.

2. Total hot length = 8 × 3 = 24 m. Dwell = 24/30 = 0.8 min = 48 s. Yes — longer than the 20 s minimum (either reduce speed is unnecessary; note the fabric must also reach temperature, so ample dwell is acceptable within scorch/sublimation limits).
3. (a) Padding mangle → stenter (dry) → curing oven/stenter hot zones → calender if specified → rubber-belt sanforizer with Palmer dryer → inspection. (b) Padding mangle → stenter dry → stenter hot zones or curing oven to cure the fluorocarbon → inspection (no sanforizer; dimensional control is by heat setting on the stenter).
4. Concentration = 12 g/kg ÷ 0.15 L/kg = 80 g/L.
5. Uneven compressive shrinkage across the width — worn/unevenly ground rubber belt or uneven belt stretching/pressure in the shrinkage zone. Check: inspect and measure the belt across its width, verify pressure settings, and re-test left/centre/right wash specimens after correction.

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